Classification of systems:

Lecture (7)

1- Linear and non-Linear system.

2- Time variant and Time Invariant sys.

3- Linear time variant (LTV) and Linear time Invariant (LTI) systems.

4- static and dynamic systems.

5. Casual and non-Casul system.

6- Invertible and non invertible sys.

7. stable and unstable systems.

1- Linear and non-Linear system.

* A system is said to be Linear if it satisfies the superposition principle. Consider a system with input $x_1(t)$, $x_2(t)$ and output $y_1(t)$ $y_2(t)$ for Linearity $T\{a_1 x_1(t) + a_2 x_2(t)]$

$$= a_1 T\{x_1(t)\} + a_2 T\{x_2(t)\}.$$

Example: $x(t) \rightarrow \boxed{\text{system}} \rightarrow y(t) = x^2(t)$

Sol:

when input $a_1 x_1(t)$ we get output $y_1(t) = a_1^2 x_1^2(t)$

" " $a_2 x_2(t)$ " " " " $y_2(t) = a_2^2 x_2^2(t)$

" " $a_1 x_1(t) + a_2 x_2(t)$ the output signal

from system $= (a_1 x_1(t) + a_2 x_2(t))^2$

the system is non-Linear because the output for sum two individual input ≠ common input.

E.X2 $x(t) \rightarrow \boxed{\text{System}} \rightarrow y(t) = x(t-2)$

Sol for input signal $a_1 x_1(t)$ output $y_1(t) = a_1 x_1(t-$

" " " $a_2 x_2(t)$ " " $y_2(t) = a_2 x_2(t-2)$

" Common input $a_1 x_1(t) + a_1 x_2(t)$

the output $a_1 x_1(t-2) + a_2 x_2(t-2)$

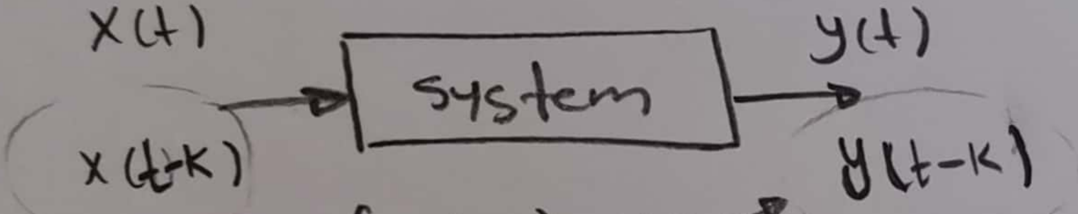
So the system is Linear.

2- Time Variant and Time Invariant system:- (3)

* A system is said to be time variant if is input; output characteristic change with time. otherwise said to be time invariant system.

* the condition for time invariant is

output with shift the input signal must be equal to shift the output.



if x(t-k) give → y(t-k)

that means:

① compute the output signal with respect to shift the input signal by k x(t-k).

② compute the output signal after shifting the time by k y(t-k).

③ if the output in frist y(n,k) = y(n-k) the system is Invariant.

Ex: $y(t) = x(t) + x(t-2)$

Sol

first $y(t,k) = x(t-k) + x(t-2-k)$

second $y(t-k) = x(t-k) + x(t-2-k)$

$\because y(t,k) = y(t-k)$ the system is Time invariant.

E.x $y(t) = x^2(t) + t\,x(t+5)$

Sol:

* $y(t,k)$ the output signal with shifting the input signal by $k \Rightarrow x(t-k)$

$y(t,k) = x^2(t-k) + t\,x(t+5-k)$.

* the output signal with shift by k $y(t-k)$ equal to $x^2(t-k) + (t-k)x(t+5-k)$

$y(t,k) \neq y(t-k)$

the system is Time invariant.

(3) Linear time variant (LTV) and Linear time Invariant (LTI) system.

④ static and Dynamic System :

static system is memory Less system.
Dynamic system is memory system.

* IN static system the output depend on the same instant input. "present time).

* IN Dynamic system the output signal depend on the past or Future time.

E.X $y(t) = 2x^2(t)$

at $t=0 \Rightarrow y(0) = 2x^2(0)$
$t=2 \Rightarrow y(2) = 2x^2(2)$

the system is static.

EX $y(t) = x(t) + x(t-1)$

at $t=0 \Rightarrow y(0) = x(0) + x(-1)$
$t=2 \Rightarrow y(2) = x(2) + x(1)$

the system is Dynamic

EX $y(t) = x(t) + x(t+5)$

$y(0) = x(0) + x(5)$

The system is Dynamic.

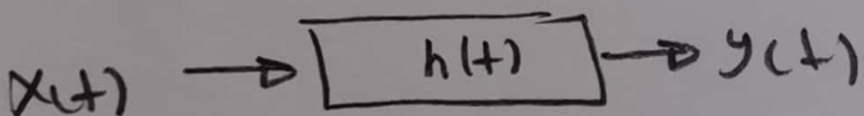
(5) Causal and non Casual systems :-

* A system is said to be Causal if it's response is depend upon Present or Past input and doesn't depends up Future input, non Causal depend upon Future.

* All non Causal systems are dynamic but versia is not true.

* All static systems are Causal but versia is not true.

(6) Invertible and non-invertible system



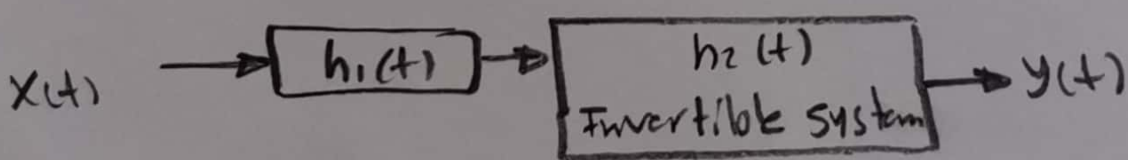
Laplace transform

$$y(t) = x(t) * h(t).$$

$$Y(s) = X(s) \cdot H(s) \Rightarrow H(s) = \frac{Y(s)}{X(s)}$$

Laplace inverse

h $h(t) =$ ✓



if $y(t) \neq x(t)$ the system is non invertible

$$H_2(s) = \frac{1}{H_1(s)}$$

⑦ stable and unstable systems :

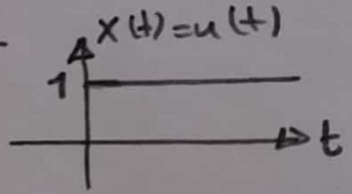
→ A system is said to be stable, when it Produce bounded outPut for Bounded input.

→ Bounded mean the Amp of signal don't equal to ∞ at any time.

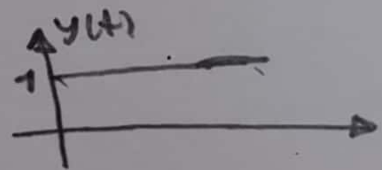
E.X test the stable of system

$$x(t) \rightarrow \boxed{\text{system}} \rightarrow y(t) = x^2(t)$$

sol
for BI



the out Put $= y(t) = u(t)$.



so the out Put is also Bounded and system is stable.

EX test the stable of system $y(t) = \int_{-\infty}^{t} x(t)\, dt$

sol for BI $x(t) = u(t)$

the $y(t) = \int_{-\infty}^{t} u(t)\, dt = r(t)$ the out is not Bounded so the system is unstable.